Date: _____

Name: M.AbdUllah

Class: BSCS

Section: D (2026to2030)

Task NO=01

Program=01.

Output=>
A
B

Program=02

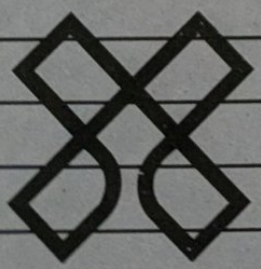
Output=>
A
B

Program=03

Output=>
66
65

Task NO=02.

```
#Include <iostream>
using namespace std;
Int main()
{
Int a, b;
```



Date:

```
Cout << " Enter your First number : 10 ";
    Cin >> a;
Coubt<< " Enter your second number : 2 ;
    Cin >> b;

Coubt<< " Addition = " << a+b << endl;
Coubt<< " Subtraction = " << a-b << endl;
Coubt<< " Multiplication = " << a*b << endl;
Coubt<< " Division = " << a/b << endl;
    return 0;
}
```

OutPut =>

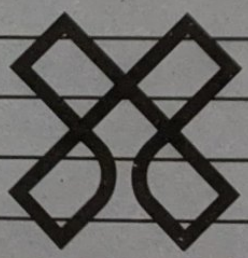
Enter your First Number 10
and Second Number 2.

Addition = 12
Subtraction = 8
Multiplication = 20
Division = 5

Task NO = 03.

OUtPUt =>

if we Puts the value 20, 22, 24
The Outs Puts will be
20 22 24
24 22 20



Amir

Date:

Task No=04

```
#Include <iostream>
#Include <cmath>
using namespace std;

Int main()
{
double x1, y1, x2, y2, d;
Coub << "Enter x1:2";
 Cin >> x1;
Coub << "Enter y1:3";
Cin >> y1;
Coub << "Enter x2:6";
 Cin >> x2;
Coub << "Enter y2:6";
 Cin >> y2;

d = Sqrt(Pow(x2-x1, 2)+
    pow(y2-y1, 2));
Coub << "Distance = " << d;
 return 0;
}
```

OutPut =>

Distance = 5

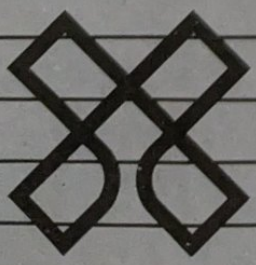
Task No = 05

1.
```
#Include <iostream>
using namespace std;
Int main()
{
    double r, area;
    cout << "Enter radius:";
    cin >> r;
    area = 3.1415'9 * r * r;
    Coute << "Area of circles=" << area;
    return 0;
}
```

OutPuts =>

if the radius is 2
the area of circle is
12.5'664

2.
```
#Include <iosfream>
using sparename std;
Int main()
    double length, width, area;
Coute << "Enter length:";
    Cin >> length;
Coute << "Enter width:";
    Cin >> width;
cout << "Enter area:";
    Cin >> area;
```



Amir

```
area = length * width;
    Cout << "Area of Rectangle=" << area;
      return 0;
    }
```

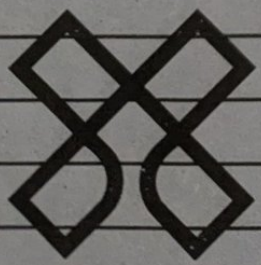
OutPut =>

length = 4
width = 6
than area of rectangle = 24

3.
```
# Include <iostream>
    using namespace std;
      Int main ()
    {
      double base, height, area;
      Cout << " Enter base: ";
       cin >> base;
       Cout << "Enter height:";
         Cin >> height;
       area = base * height * 0.5;
        Cout << "Area of triangle = " << area;
         return 0;
```

OutPut =>

base = 5
height = 6
AOT = 15



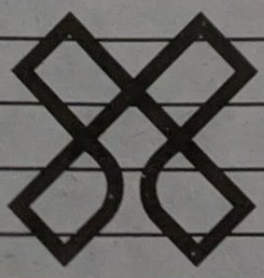
Amit

Task NO= 06.

```
#Include <iostream>
using namespace std;
Int main ()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;
    double Subtotal = item1 + item2 + item3
                    + item4 + item5
    double tax = Subtotal * 0.06;
    double total = Subtotal + tax;

    Cout << "Price of item 1 = $" << item1 << endl;
    Cout << "Price of item2 = $" << item2 << endl;
    Cout << "Price of item3 = $" << item3 << endl;
    Cout << "Price of item4 = $" << item4 << endl;
    Cout << "Price of item5 = $" << item5 << endl;

    Cout << "Subtotal = $" << Subtotal << endl;
    Cout << "Sales Tax = $" << tax << endl;
    Cout << "Total = $" << total << endl;
    return 0;
}
```



Amir

Name: M Abdullah **Section: BSCS (D)**
Roll No: 44 **Program: Lab 2**

Program 1: Arithmetic Operations (Code & Output)

Code:

```cpp
#include <iostream>
using namespace std;

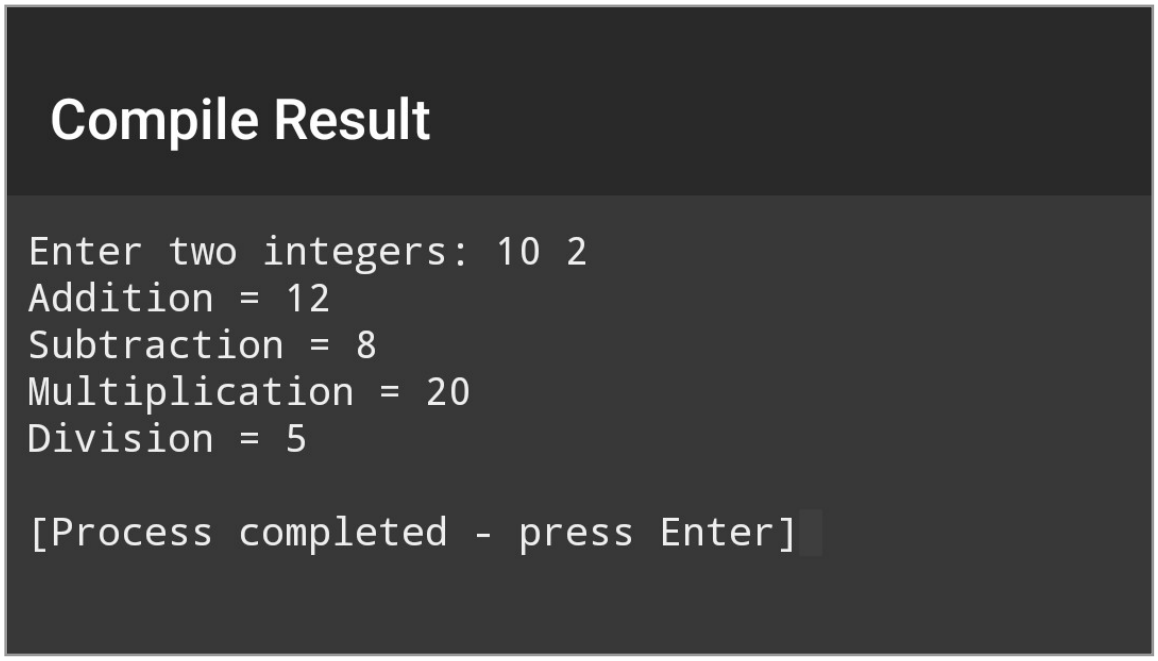
int main()
{
    int a, b;

    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Addition = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    cout << "Division = " << a / b << endl;

    return 0;
}
```

Output:



Name: M Abdullah **Section: BSCS (D)**
Roll No: 44 **Program: Lab 2**

Program 2: Area of Circle (Code & Output)

Code:

Coding C++
Auto saved at 21:11:00 RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    double r, area;

    cout << "Enter radius:";
    cin >> r;

    area = 3.14159 * r * r;

    cout << "Area of Circle = " << area;

    return 0;
}
```

Output:

Compile Result

```
Enter radius:2
Area of Circle = 12.5664
[Process completed - press Enter]
```

Name: M Abdullah **Section: BSCS (D)**
Roll No: 44 **Program: Lab 2**

Program 3: Area of Rectangle (Code & Output)

Code:

Coding C++
Auto saved at 21:19:21 RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    double length, width, area;

    cout << "Enter length: ";
    cin >> length;

    cout << "Enter width: ";
    cin >> width;

    area = length * width;

    cout << "Area of Rectangle = " << area;

    return 0;
}
```

Output:

Compile Result

```
Enter length: 4
Enter width: 6
Area of Rectangle = 24
[Process completed - press Enter]
```

Name: M Abdullah **Section: BSCS (D)**
Roll No: 44 **Program: Lab 2**

Program 4: Area of Triangle (Code & Output)

Code:

Coding C++
Auto saved at 21:21:14
RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    double base, height, area;

    cout << "Enter base: ";
    cin >> base;

    cout << "Enter height: ";
    cin >> height;

    area = base * height * 0.5;

    cout << "Area of Triangle = " << area;

    return 0;
}
```

Output:

Compile Result

```
Enter base: 5
Enter height: 6
Area of Triangle = 15
[Process completed - press Enter]
```

Name: M Abdullah **Section: BSCS (D)**
Roll No: 44 **Program: Lab 2**

Program 5: Distance Between Two Points (Code & Output)

Code:

Coding C++
Nothing changed
RUN MENU

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2, d;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    d = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

    cout << "Distance = " << d;

    return 0;
}
```

Output:

Compile Result

```
Enter x1: 2
Enter y1: 3
Enter x2: 6
Enter y2: 6
Distance = 5
[Process completed - press Enter]
```

Name: M Abdullah **Section: BSCS (D)**
Roll No: 44 **Program: Lab 2**

Program 6: Separate Digits of a Five-Digit Number (Code & Output)

Code:

Coding C++
Auto saved at 21:54:17
RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a five-digit number: ";
    cin >> num;

    cout << num / 10000 << "   ";
    cout << (num / 1000) % 10 << "   ";
    cout << (num / 100) % 10 << "   ";
    cout << (num / 10) % 10 << "   ";
    cout << num % 10;

    return 0;
}
```

Output:

Compile Result

```
Enter a five-digit number: 12345
1   2   3   4   5
[Process completed - press Enter]
```

Name: M Abdullah **Section: BSCS (D)**
Roll No: 44 **Program: Lab 2**

Program 7: Bill with Sales Tax (Code & Output)

Code:

Coding C++
Auto saved at 21:41:18
RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal = item1 + item2 + item3 + item4 + item5;
    double tax = subtotal * 0.06;
    double total = subtotal + tax;

    cout << "Price of item 1 = $" << item1 << endl;
    cout << "Price of item 2 = $" << item2 << endl;
    cout << "Price of item 3 = $" << item3 << endl;
    cout << "Price of item 4 = $" << item4 << endl;
    cout << "Price of item 5 = $" << item5 << endl;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << tax << endl;
    cout << "Total = $" << total << endl;

    return 0;
}
```

Output:

Compile Result

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax = $3.825
Total = $67.575

[Process completed - press Enter]
```